CHAPTER 3

INSTALLATION

Peerless
DOUGH MIXING & MAKE-UP
A Peerless Group Company

CHAPTER 3
TABLE OF CONTENTS

INSTALLATION

CHAPTER 3
LIST OF FIGURES, CHARTS, TABLES, EXAMPLES

INSTALLATION

SECTION 3-1
OFF LOADING AND HOISTING

⚠ **WARNING**	Your mixer is a heavy, awkward machine. If tipped over, mixer could cause serious injury or death. Use only qualified and experienced riggers to move and place mixer.

NOTE:	If you have any questions about this procedure, call the Service Department at **Peerless Machinery Corp.**

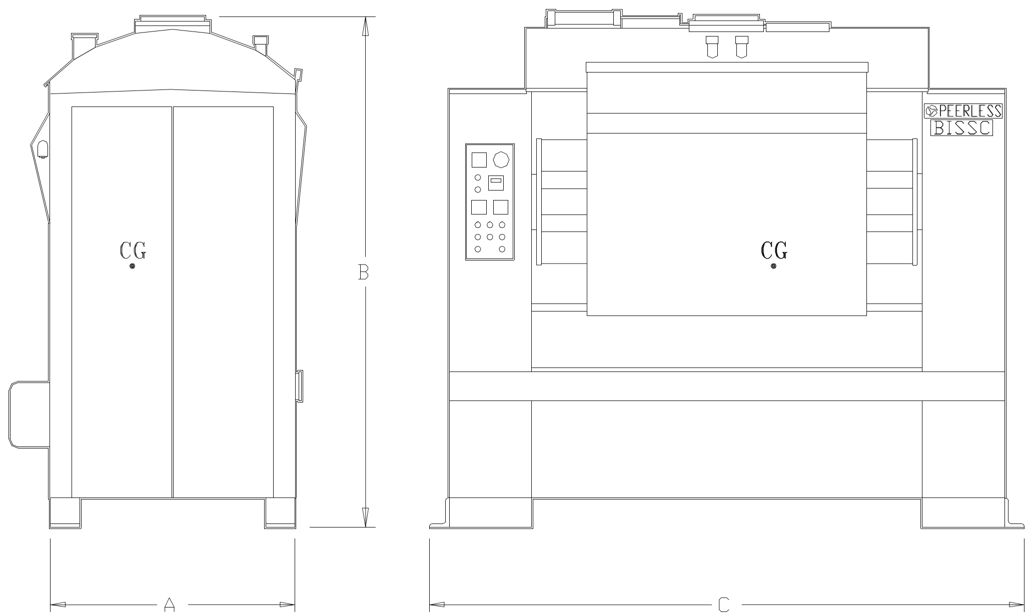
Your **Peerless** mixer will be enclosed in protective shrinkwrap on a wooden skid. If your mixer has been shipped overseas, it will come to you in a totally enclosed crate on a wooden skid.

The weight of your mixer depends on the model number you have ordered. See **Figure 3-1-1, Mixer Specifications**, for approximate mixer weights and overall dimensions. See Certified Data Sheet in this manual for actual mixer dimensions. The model number of your mixer is given on the cover page of the manual and in **Chapter 2, Options Checklist.**

NOTE:	Do not lift mixers using canopy (bowl cover), agitator shaft, or bowl assembly. *Lift mixers only at specified lift points.*

SECTION 3-1

MIXER SPECIFICATIONS FIGURE 3-1-1



CG = Approximate center of gravity location of mixer

MODEL NUMBER	SHIPPING WEIGHT LBS. (KGS)	NET WEIGHT LBS (KGS)	FLOOR LOAD LBS/FT² (KGS/CM²)	A FT-IN (M)	B FT-IN (M)	C FT-IN (M)
DA100	10,400 (4,717)	10,000 (4,535)	5,520 (2.5)	3'6" (1.07)	7'½" (2.15)	6'6" (1.99)
DA150	10,800 (4,899)	10,400 (4,717)	6,190 (2.7)	3'6" (1.07)	7'½" (2.15)	7'1" (2.17)
DA200	11,900 (5,398)	11,400 (5,171)	4,420 (2.2)	3'6" (1.07)	7'8" (2.34)	9'4" (2.89)
DA250	12,400 (5,625)	11,900 (5,398)	4,660 (2.3)	3'6" (1.07)	7'10" (2.39)	9'4" (2.89)
DA300	13,100 (5,942)	12,550 (5,693)	5,010 (2.4)	3'10" (1.17)	8'4" (2.55)	10'0" (3.06)
DA350	14,200 (6,441)	13,650 (6,192)	5,490 (2.7)	3'10" (1.17)	8'4" (2.55)	10'0" (3.06)
DA400	21,800 (9,888)	21,100 (9,571)	4,990 (3.6)	4'8" (1.42)	9'5 ½" (2.89)	10'0" (3.06)
DA500	23,300 (10,569)	22,550 (10,229)	5,440 (3.9)	4'8" (1.42)	9'5 ½" (2.89)	10'8" (3.26)
DA600	25,400 (11,521)	24,600 (11,159)	6,020 (4.4)	4'10" (1.47)	9'7 ½" (2.94)	11'3" (3.43)
DA650	26,000 (11,793)	25,200 (11,430)	6,170 (4.5)	4'10" (1.47)	9'7 ½" (2.94)	11'3" (3.43)

SECTION 3-1

Peerless mixers shipped on wooden skids can be removed from the truck in one of three ways:

FORKLIFT.

Lift the mixer using skids provided.

On smaller mixers, place the forks under front and rear skids.

On larger mixers, a special steel front skid is used due to the cutout in the front of the mixer. Place the forks through the notch in the steel beam and through the notch in the rear wooden beam. See **Figure 3-1-3**.

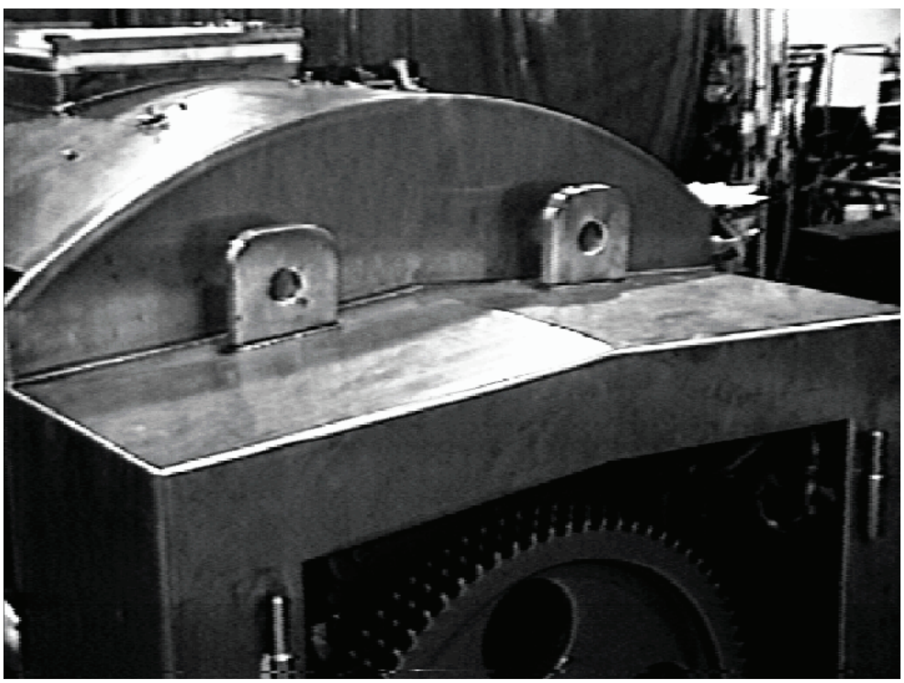
NOTE: It is best to lift mixer from rear if possible.

CRANE.

Lift the mixer using a crane by attaching straps around the mixer and under the skids. Take care to use spreader bars as needed so straps do not damage the mixer.

Optional lifting lugs can be used to lift the mixer. See **Figure 3-1-2 Optional Lifting Lugs**. *Lifting lugs are optional and are supplied only on request.*

OPTIONAL LIFTING LUGS FIGURE 3-1-2



DRAGGING. You can drag or push the mixer while it is still mounted to the skid by using a forklift to pick up one end of the mixer. Take care to protect the mixer's finish by using wooden blocks between the forklift and the mixer end.

> **NOTE:** On the larger mixer line, the mixer holddown feet are removable. The wooden rear skid is attached to the mixer by the rear mixer holddown feet. After the skid is removed, remove holddown feet and rotate 90 degrees so the angle is against the floor. Bolt back to the leg using the same screws. The holddown feet for the front legs are shipped loose.

Peerless mixers with export crating can be lifted in two ways:

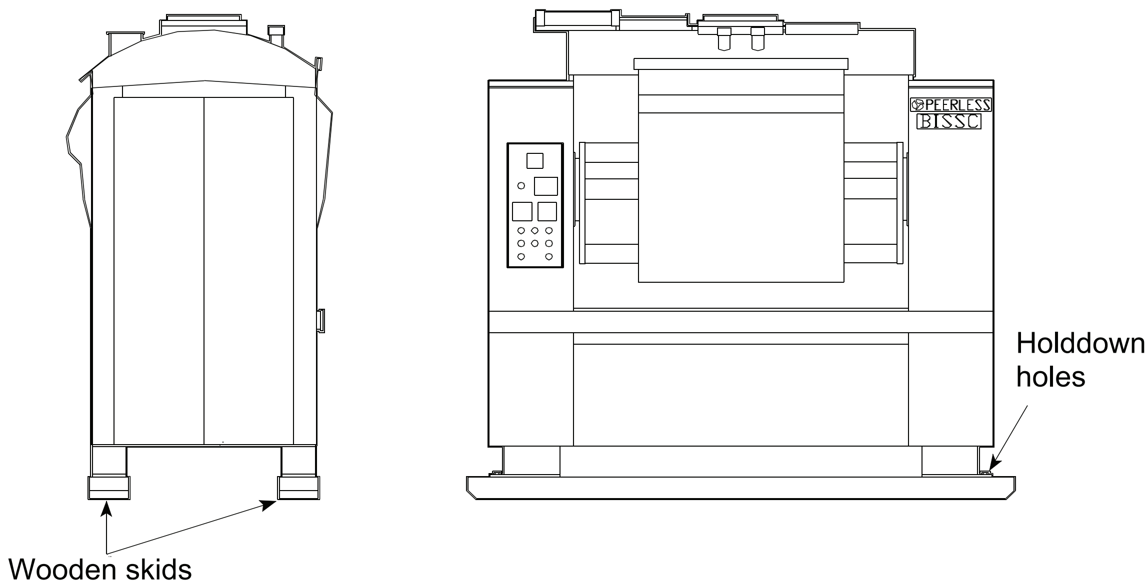
FORKLIFT. Place the forks under the skids.

CRANE. Attach straps around the crating. Use spreader bars so the crate is not damaged.

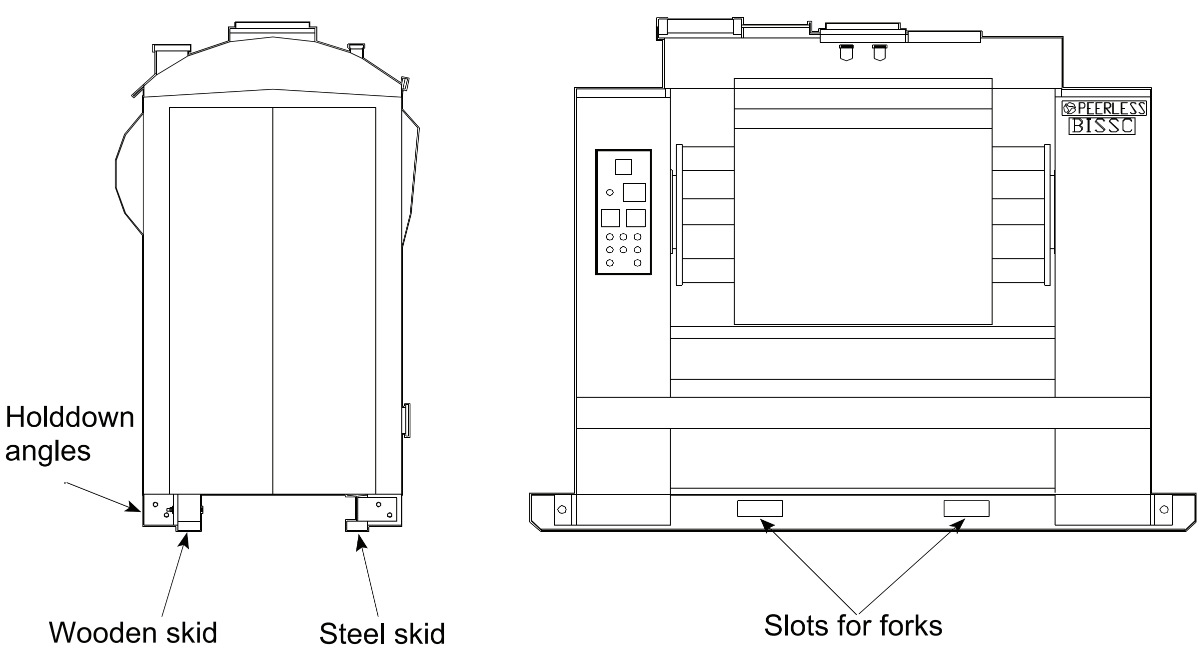
Once the mixer is uncrated, it can be lifted and moved as noted before.

SECTION 3-1

MIXER SHIPPING SKIDS ARRANGEMENT FIGURE 3-1-3



SMALL MIXERS: DA100, DA150



LARGE MIXERS: DA200-DA650

SECTION 3-2
UNPACKING

Your **Peerless** mixer will arrive enclosed in shrinkwrap on a wooden skid. If it has been shipped overseas, the shrinkwrapped mixer will come in a solid waterproof crate on a wooden skid. All components shipped loose, such as coolant hoses, electrical enclosure, hose clamps, fittings, etc., are shipped in a separate crate.

Remove the lumber crate or shrinkwrap from the skid. Do not remove the skid itself until the mixer has been moved to its permanent location. Store the crate containing the electrical enclosure, hoses and fittings in a safe place so it is out of the way until the mixer has been set in place.

All **Peerless** mixers are inspected and tested at the factory before shipment. Have the installer carefully reinspect your mixer for loose, damaged, or broken parts. Report any damages immediately to the carrier.

A packing list has been attached to the crate that holds the electrical enclosure. If your mixer does not have a electrical enclosure, the packing list will be attached to the mixer end door. Most loose parts will be on the skid with the electrical enclosure. Some parts are shipped inside the end door of the mixer. Report any shortage of parts to the Parts Department at **Peerless Machinery Corp.**

SECTION 3-3
SITE PREPARATION

The standard **Peerless** dough mixer is designed with four rectangular legs. These legs raise the base of the mixer 6" (15cm) off the floor to allow for cleaning under the mixer. Optional leg heights and configurations are available. See the **Certified Data Sheet** and **Chapter 2, Options Checklist** in this manual to see if your mixer has optional leg heights.

When deciding where the mixer will be located, consider the following points:

- ✔ Provide ample space on all four sides of the mixer for cleaning and maintenance. We recommend a minimum clearance of 3' (1m) on the ends and rear, and 5' (1.5m) at the front.
- ✔ If possible, allow room at either end of the mixer for pulling the agitator shaft. Allow about 6" (15cm) less than the overall length of the mixer. See **Certified Data Sheet** in this manual.
- ✔ Allow enough room behind the mixer for the refrigeration hose loop and for removal of the motor. Refer to **Figure 3-6-1** for refrigeration hose connections, and to the **Certified Data Sheet** for the dimension required to remove the motor.
- ✔ The end doors can be lifted off if necessary.
- ✔ The drive end of the mixer (sprockets, chain and drip oiler) is on the right end of the mixer.
- ✔ The **hydraulic tilt system** is on the left-hand end of the mixer.
- ✔ Install the mixer near a waste-water drain.
- ✔ Install the electrical enclosure in a dry place near the mixer and in direct sight of the operator. Refer to **Section 3-5**.

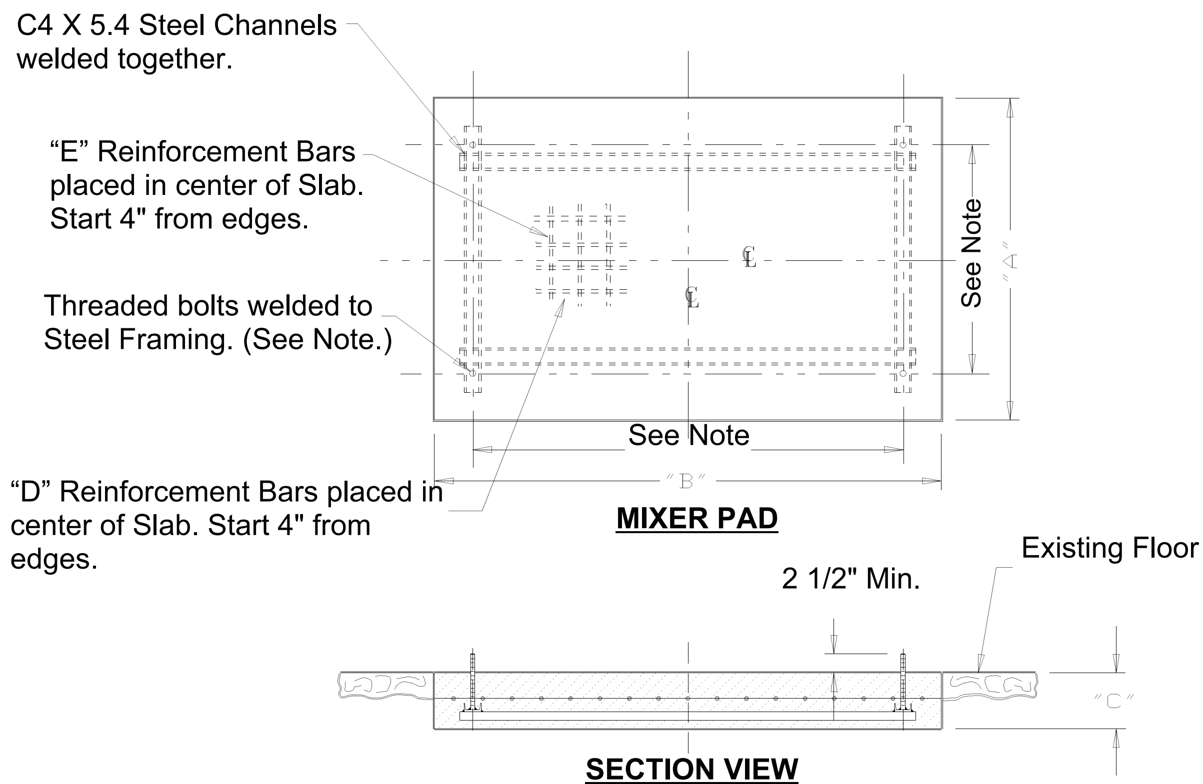
SECTION 3-3

When a **Peerless** dough mixer is running, the load is such that the mixer must be mounted to the floor or pad. The **Certified Data Sheet** shows the actual style of the mixer hold-down legs and the dimension and location of the mounting bolts. See **Figure 3-1-1** for mixer weight, floor load, shock load, and center of gravity location for your mixer.

Because of the many factors in designing a pad for the mixer, we strongly recommend that an architect or structural engineer be consulted. As a general guideline, however, **Figure 3-3-1** shows a recommended mixer foundation pad for a typical application.

SECTION 3-3

RECOMMENDED MIXER FOUNDATION PAD FIGURE 3-3-1



NOTE: See Certified Data Sheet for actual bolt locations.

MODEL	"A" FT-IN (M)	"B" FT-IN (M)	"C" IN (CM)	"D"	"E"
DA100	7'0" (2.13)	9'10" (3.00)	6" (15.24)	6-#5	9-#5
DA150	7'0" (2.13)	10'8" (3.25)	6" (15.24)	7-#5	10-#5
DA200	7'0" (2.13)	13'2" (3.96)	6" (15.24)	7-#5	12-#5
DA250	7'0" (2.13)	13'2" (3.96)	6" (15.24)	7-#5	12-#5
DA300	7'6" (2.29)	14'0" (4.27)	6" (15.24)	7-#5	12-#5
DA350	7'6" (2.29)	14'0" (4.27)	6" (15.24)	7-#5	12-#5
DA400	8'0" (2.44)	15'0" (4.57)	8" (20.32)	8-#5	14-#5
DA500	8'4" (2.54)	14'6" (4.42)	8" (20.32)	7-#5	14-#6
DA600	8'4" (2.54)	15'2" (4.62)	9" (22.86)	8-#6	15-#6
DA650	8'4" (2.54)	15'2" (4.62)	9" (22.86)	8-#6	15-#6

SECTION 3-4
MIXER INSTALLATION

Move your **Peerless** mixer to the prepared site. Bolt the mixer to the floor using the hold-down holes provided. See the **Certified Data Sheet** for size and location.

To help minimize vibration, a damping material can be placed under the legs. **Vibration Isolation Pads** (Vibro Pads) are optional and can be purchased at the time of order. **Peerless** uses 1/2" (1.25 cm) thick neoprene **Vibration Isolation Pads** made by **Fabcel, P/N #300.** The load rating is 200 to 300 PSI (13.8-20.6 bar).

In some situations, mixer legs must be removable for shipping. See the **Certified Data Sheet** or **Chapter 2, Options Checklist,** to see if your mixer's legs are removable. There are two types of legs: short two-bolt legs and tall four- or six-bolt legs. See **Figure 3-4-1, Mixer Bolt-On Legs.** Removable legs must be bolted to the base plate before the mixer is installed. Before assembling, find your mixer's serial number, a dash, and the leg number on each leg (ex. 95004-1). Match numbers stamped on the base (1, 2, 3, 4) with numbers stamped on the legs. Attach the legs to your mixer using the bolts provided.

Level the mixer from side to side by laying a level on the agitator shaft. Level the mixer from back to front by placing the level vertically on the inside corner of the column frame--check both sides and use the average. The base plate at the rear of the mixer can also be used to check for level.

Once the mixer has been leveled and anchored to the floor, make the following connections. Refer to the manual section indicated below for details on making these connections.

1. Wire the mixer to the remote-mounted electrical enclosure and related loose components. See **Section 3-5, Mixer Wiring**.

2. Attach the refrigeration hoses and mount the related components. See **Section 3-6, Refrigeration Connection--Indirect Liquid**.

SECTION 3-4

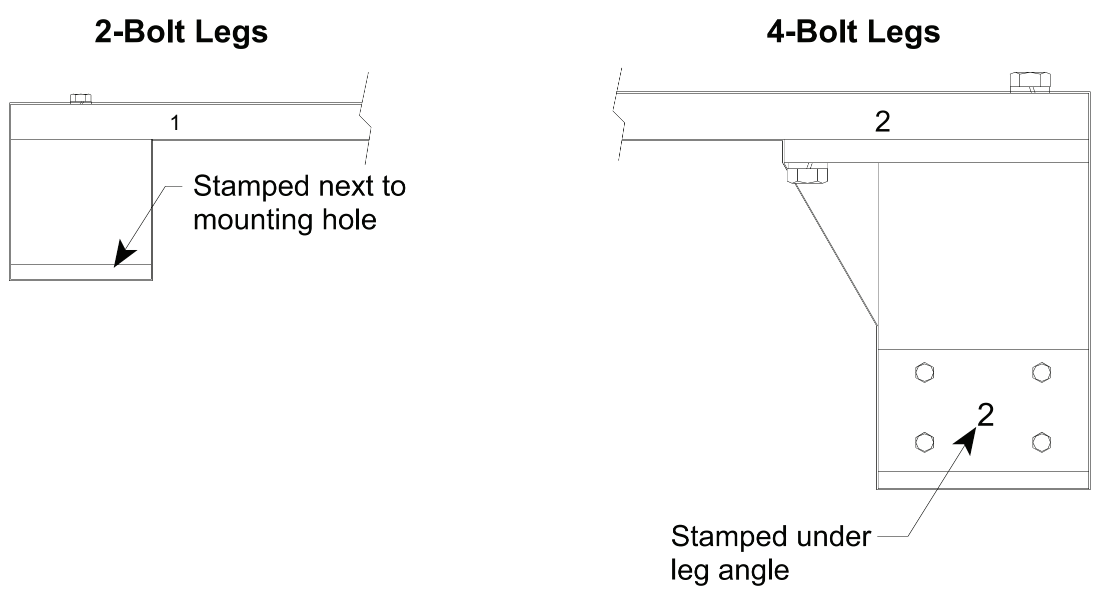
3. Attach the flour hopper connection. See **Section 3-8, Flour Gate Connection**.

4. Attach the air supply. See **Section 3-9, Air Supply Connection**.

5. Make liquid ingredient line connections. See **Section 3-10, Liquid Ingredient Connections**.

6. Install the **Dust Vent**. See **Section 3-11, Flour Dust Vent**.

MIXER BOLT-ON LEGS FIGURE 3-4-1



SECTION 3-5
MIXER WIRING

⚠ **CAUTION**	Read **Chapter 1, Safety** before attempting to wire mixer.

Peerless mixers are supplied with a standard remote wall-mounted electrical enclosure. The mixer's main power disconnect, circuit breaker, motor starters, **PLC** and control transformer are mounted in this electrical enclosure.

Locate the electrical enclosure; wall-mount and wire it to the mixer during mixer installation following the procedure below. If you have any questions about this procedure, call the Service Department at **Peerless Machinery, Corp.**

⚠ **DANGER**	High voltage electrical shock possible. Disconnect and lockout branch circuit feeder before wiring electrical enclosure.

⚠ **DANGER**	High voltage electrical shock and/or unexpected machine operation possible due to water shorting electrical circuits. All electrical connections *must* be sealed to prevent water from entering.

⚠ **CAUTION**	Enclosure and mixer wiring should be done by qualified and experienced electricians only.

Follow the **National Electrical Code (NEC)** and all local codes when mounting the electrical enclosure and wiring the mixer.

You and your electrical contractor should determine the location of the electrical enclosure. We strongly suggest that the enclosure be located in direct sight of the mixer so the mixer can be seen when the power is turned on. In addition, the electrical enclosure should be easily reachable by the mixer operator. Install the electrical enclosure where it will stay dry when cleaning the mixer and surrounding area. The **Peerless** standard electrical enclosure has a NEMA 12 rating--it is dust tight but not water tight.

An optional, water-tight NEMA 4 or NEMA 4X electrical enclosure can be supplied. However, we strongly recommend it also be located where it will stay dry during mixer cleaning.

The circuit breaker on your **Peerless** mixer was sized for a heavy duty jogging application. It is sized per **NEC** Article 430-52(a), exception 1 at maximum 250% drive motor FLA. See **Figure 3-5-1, Mixer Field Wiring Diagram,** for the breaker size used. Size the branch circuit feeder protection accordingly.

See **Figure 3-5-1** and mixer wiring diagrams included in this manual before wiring the mixer.

Figure 3-5-1 shows the recommended wire size required for the feeder circuit and drive motor wires. The size listed is per Article 430-22(a) of the **NEC**. These sizes should be checked against local codes by a qualified electrician.

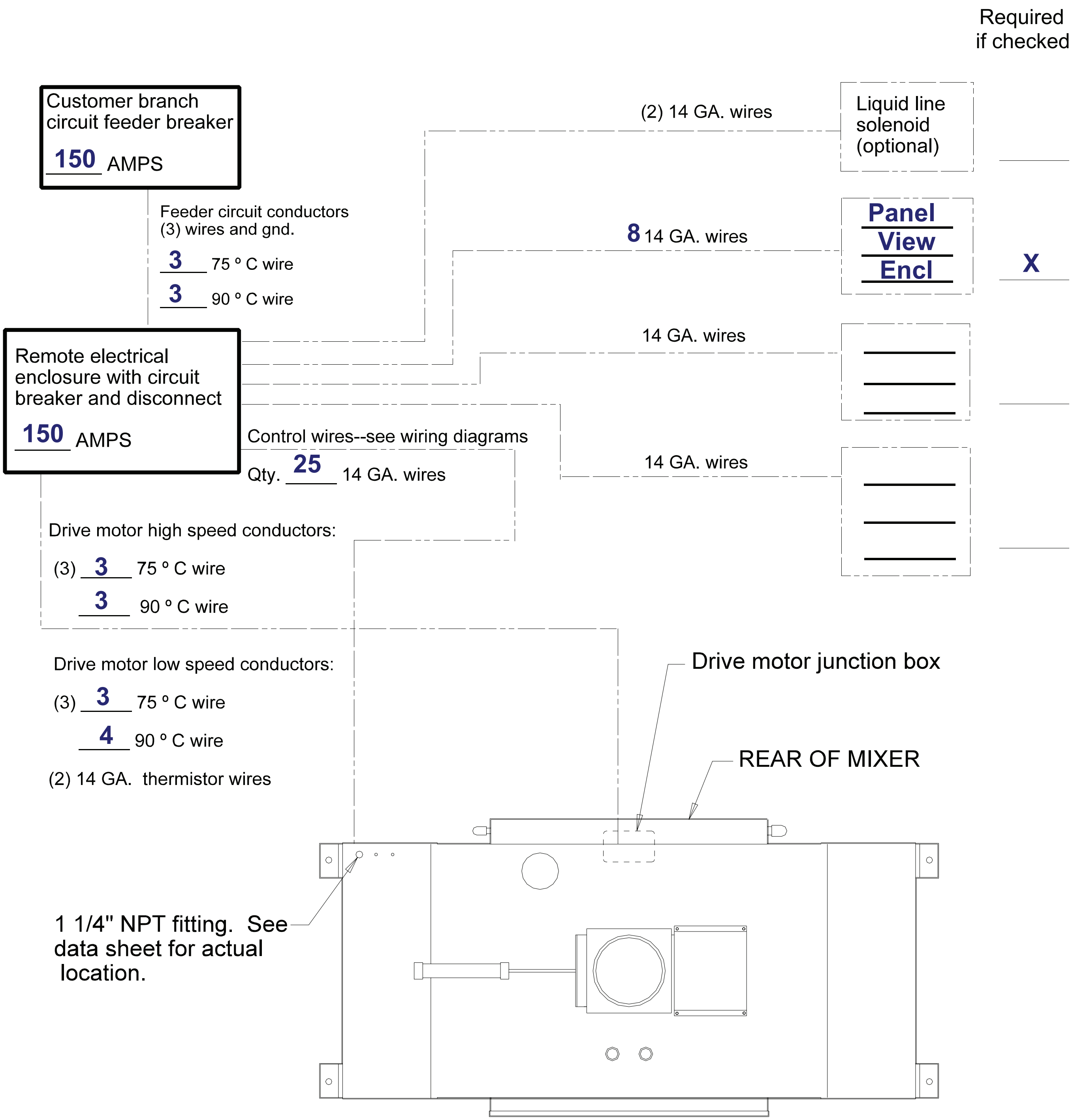
The standard drive motor for **Peerless** mixers is a two-speed, constant-torque motor. This motor requires six wires run from the drive-motor starter in the electrical enclosure to the motor junction box mounted on the motor.

Chart 3-5-2, **Mixer Field Wiring Summary,** lists the devices and conduit runs that need to be made. This list includes the most common options that can be supplied. Refer to the wiring diagrams for your mixer and **Chapter 2, Options Checklist,** for options included on your mixer.

See **Figure 3-5-3** for an example of how to wire the electrical enclosure terminal strip to the mixer terminal strip. The mixer terminal strip is located at the rear of the mixer on the column behind the push-button panel. Remove the bolt-on cover to reach it.

SECTION 3-5

MIXER FIELD WIRING DIAGRAM FIGURE 3-5-1



TOP VIEW OF MIXER

SERIAL NO. 201031

MODEL NO. DA150FD+

SECTION 3-5

MIXER FIELD WIRING SUMMARY CHART 3-5-2

FROM	TO	DEVICE	# WIRES	WIRE SIZE
Branch circuit feeder	Electrical Enclosure	Circuit Breaker	3 & Gnd	See Note 1
Electrical Enclosure Terminal Strip (TS)	Mixer Terminal Strip (TS)	Mixer control wiring	See Note 2	#14 ga
Electrical Enclosure starter lugs and TS	Drive motor & thermistors	Drive motor & thermistors	6 motor & 2 thermistors	See Note 1 #14 ga for thermistor
Electrical Enclosure TS or Mixer TS	Liquid line solenoid	Liquid line solenoid	2	#14 ga

NOTES:

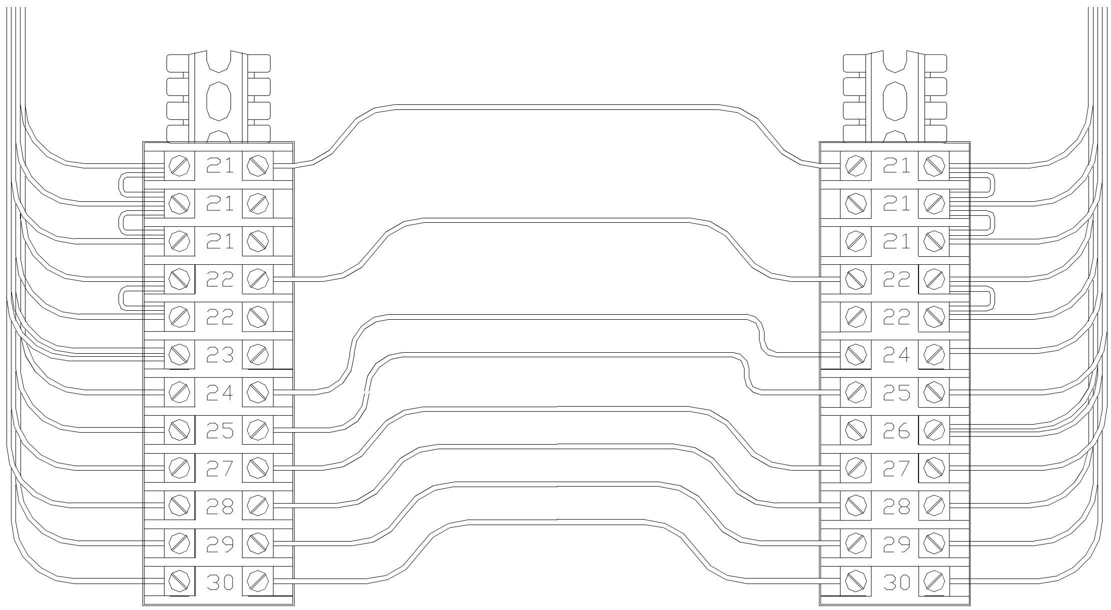
1) Wire size is determined by drive motor full load amps per the **NEC** and local codes. See **Figure 3-5-1.**

2) See **Figure 3-5-1** and mixer wiring diagrams for the number of control wires.

3) TS = Terminal Strip.

SECTION 3-5

TERMINAL STRIP WIRING EXAMPLE FIGURE 3-5-3



Electrical Enclosure Terminal Strip
(Mounted in remote electrical enclosure)

Mixer Terminal Strip
(Located in rear of mixer behind bolt-on cover)

NOTE: Terminal numbers are examples only. See wiring diagrams for actual numbers. Wire the electrical enclosure terminal strip to the mixer terminal strip by connecting the terminal in the electrical enclosure to the terminal on the mixer.

SECTION 3-5

The motor run circuit for your **Peerless** mixer is a 3-phase system. Before running, check with a voltmeter to make sure the correct voltage is present on all three phases on the feed side of the circuit breaker. See the electrical specification tag attached to the electrical enclosure lid for the voltage and frequency your mixer has been designed for.

⚠ **CAUTION**	Before turning on power to the mixer for the first time, check all terminal screws and lugs to make sure they are tight.

On most mixers, the tilt hydraulic pump motor and control transformer are made to accept multiple voltages. Both devices are wired to the voltage specifications of their particular mixer before shipping. It is a good idea, however, to check both the tilt hydraulic pump motor and control transformer wiring to be sure they are wired for the correct voltages.

After connecting power from the branch circuit feeder, check the direction of rotation of the mixing agitators and the hydraulic pump motor. Use the following procedure as a guideline.

TESTING FOR CORRECT MOTOR PHASING

1. Close and latch mixer end doors.

2. Close and latch the electrical enclosure door.

3. Turn on the mixer's power using the disconnect switch on the electrical enclosure.

4. Push the button on the front operator panel labeled **DOWN**. At the same time, push the button labeled **SAFETY**. Hold both buttons in.

5. The mixer bowl should start to tilt down.

6. If the bowl does not tilt down, do the following:

a) Open the mixer's left-hand end door to see what direction the pump motor is rotating. The motor should be rotating clockwise when you are looking at the motor fan from above.

b) If the pump motor is rotating counter-clockwise, then mixer phasing is wrong. Disconnect power to the mixer circuit breaker and switch two of three feed lines on top of the circuit breaker. Start over at step 1.

c) If the pump motor is rotating clockwise, refer to **Chapter 5, Maintenance**, for trouble-shooting.

d) If the motor does not run, refer to **Chapter 5, Maintenance**, for trouble shooting.

7. If the bowl does tilt down, mixer phasing is correct. Direction of the mixing agitators, however, must also be checked.

8. Tilt the bowl down far enough to see the mixing agitators.

9. Check inside of the bowl for any material or debris. If present, disconnect and lock out power before removing.

10. Open the right-hand end door. Check the drive chain or belt for any material or debris; disconnect and lock out power before removing. Close and latch the end door.

11. Switch the **JOG SELECTOR** switch to LOW.

12. Jog the agitators by pushing the button labeled **JOG** and at the same time, the button labeled **SAFETY**.

13. The agitators should start to rotate. If phased correctly, the front agitator will rotate in a direction that goes up toward the front of the mixer and down toward the back. See **Figure 3-5-4**. If agitator does not rotate, see **Chapter 5, Maintenance**, for trouble shooting.

a) If the agitators rotate in the opposite direction, change drive-motor phasing.

b) Disconnect and lockout power to the mixer using the disconnect on the electrical enclosure.

c) Switch two of three wires feeding the top of the drive motor starter.

d) Shut the electrical enclosure door and turn on power.

e) Start again at step 12.

14. Switch the **JOG SELECTOR** switch to HIGH.

15. Repeat step 12.

16. The agitators should change smoothly from low speed to high speed *without changing direction*.

a) If the agitators *do* change direction, change the drive motor phasing.

b) Disconnect and lockout power to the mixer using the disconnect on the electrical enclosure.

c) Switch two of the three motor lead wires connected to the high speed starter.

d) Shut the electrical enclosure door.

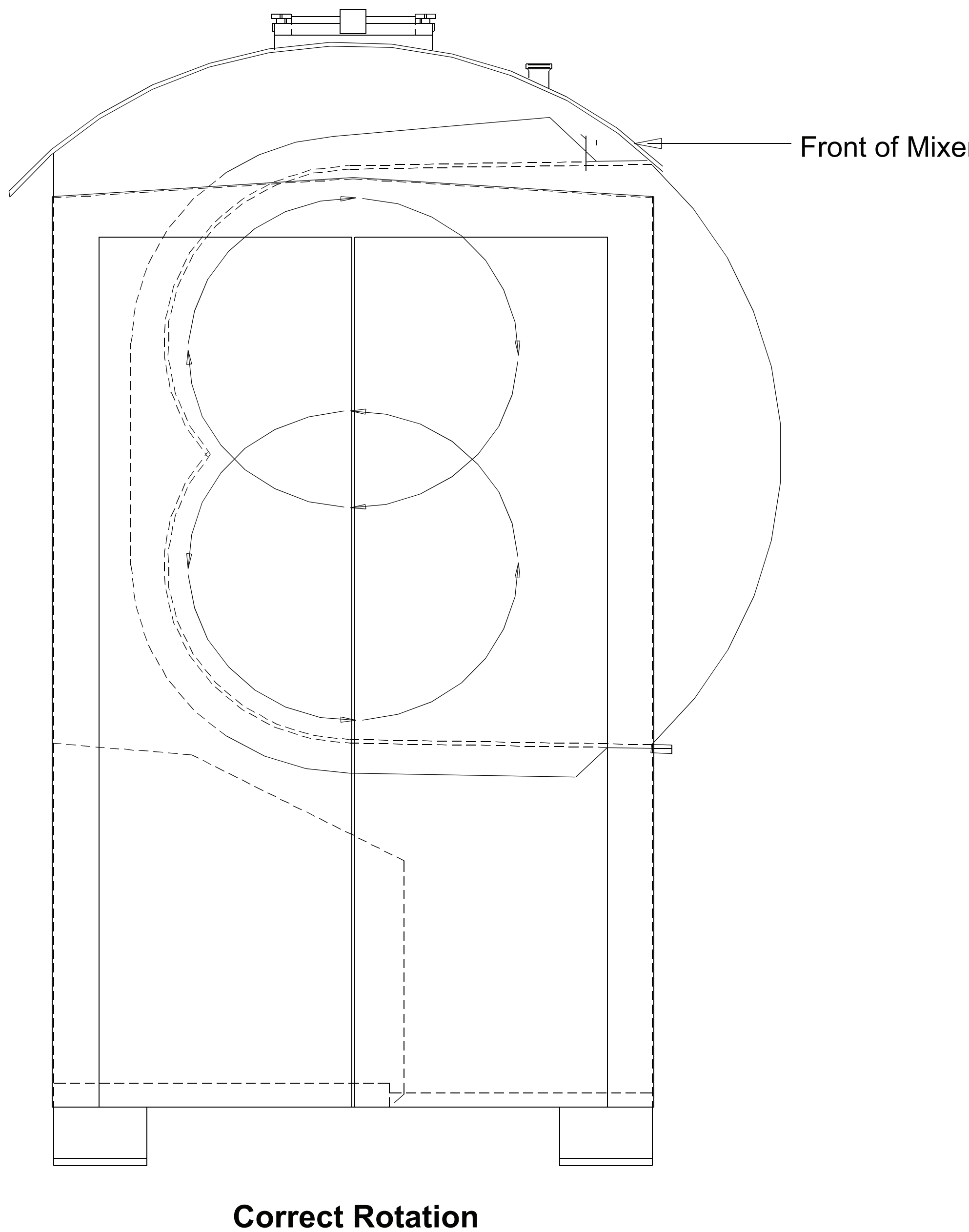
e) Repeat step 12.

The mixer is now wired.

NOTE:	Before operating mixer, read and understand **Chapter 1, Safety,** and **Chapter 4, Operation**.

SECTION 3-5

ROTATION DIRECTION OF AGITATORS FIGURE 3-5-4



Correct Rotation

SECTION 3-6
REFRIGERATION CONNECTION--INDIRECT LIQUID

WARNING	Rupturing jacket or hoses can cause serious personal injury or death. Do not exceed a jacket pressure of 235 PSIG (16.2 bar).

WARNING	Liquid trapped in the jacket can cause very high pressures. Do not install a shut-off valve in both coolant lines.

CAUTION	Correct installation of your refrigeration system is required to insure trouble-free service. We highly recommend that professional refrigeration personnel be used.

This section covers the connection of your **Peerless** mixer to an **indirect liquid refrigeration** system. Refer to **Chapter 2, Options Checklist,** to see if your mixer has this option.

An **indirect liquid refrigeration** system typically uses hot or cold water, brine or glycol as the cooling medium. *The mixer's liquid jacket is **not** designed for steam.* Connect the mixer's liquid jacket to the coolant source. Hoses are supplied with the mixer to make this connection. Install the hoses so they will not twist or kink when the bowl is tilted. The hoses should hang from the fittings on the bowl at the rear of the mixer. See **Figure 3-6-1**.

Connect the hoses to the bowl fittings and tilt the bowl. To determine the best location of the fixed end, hold the other end of the hoses feeling for pulling while the bowl tilts. The bowl should tilt without pulling or kinking the hoses.

The **refrigeration On/Off control** is optional on Double Sigma mixers. See **Chapter 2, Options Checklist**, to find out if your mixer has refrigeration On/Off control.

SECTION 3-6

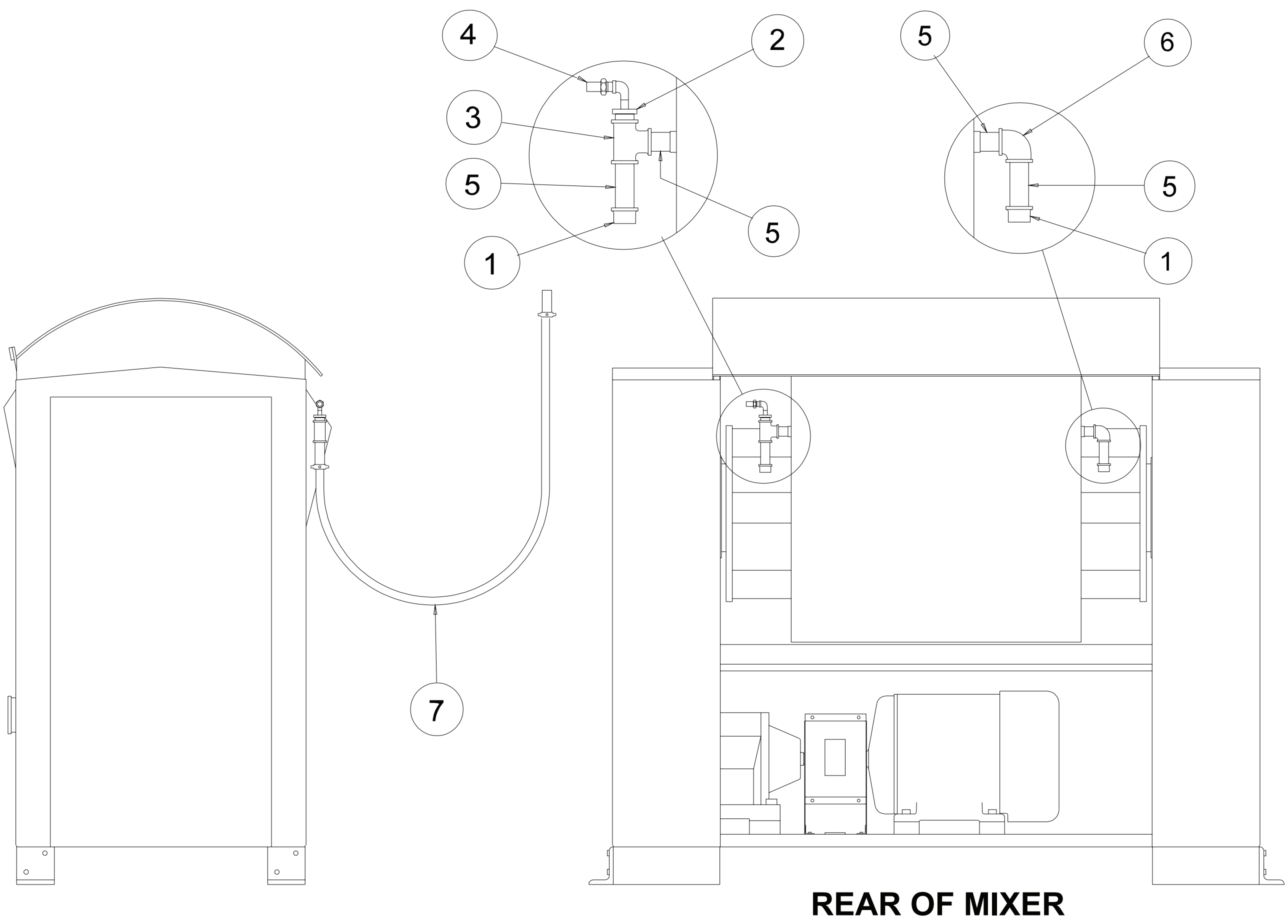
Use the optional **refrigeration On/Off control** to turn the liquid circulating pump on or off, or to open or close a valve. A voltage-free contact (dry-contact) closure controlled by the mixer run cycle is used for either way. See the mixer's wiring diagrams for an example of how to wire this control.

When the **refrigeration On/Off control** is not supplied, you or your refrigeration contractor must provide the on/off method. See **Chapter 7, Refrigeration Systems,** in this manual for more information.

See **Table 3-6-2** for hose and fitting sizes supplied for all **Peerless** mixers.

SECTION 3-6

REFRIGERATION HOSE LOCATIONS FIGURE 3-6-1



ITEM	QTY	DESCRIPTION
1	2	Galvanized Pipe Cap
2	1	Galvanized Reducer Bushing
3	1	Pipe Tee
4	1	Pressure Relief Valve
5	4	Pipe Nipple
6	1	90º Pipe Elbow
7	2	Refrigeration Hose

SECTION 3-6

INDIRECT LIQUID JACKET CONNECTION DATA TABLE 3-6-2

MIXER SIZE	WITHOUT BEC			WITH BEC		
	HOSE SIZE	FITTING SIZE	FLOW RATE	HOSE SIZE	FITTING SIZE	FLOW RATE
DA100 **DA150**	1-1/4"	1-1/4"	40-60	1-1/4"	1-1/4"	40-60
DA200 **DA250**	1-1/4"	1-1/4"	50-70	1-1/4"	1-1/4"	50-70
DA300 **DA350** **DA400**	1-1/4"	1-1/4"	60-80	1-1/2"	1-1/2"	60-80
DA500 **DA600** **DA650**	1-1/2"	1-1/2"	70-90	1-1/2"	1-1/2"	70-90

NOTES:

1) BEC = Bowl End Cooling.
2) Hose Size is given in standard SAE Hose.
3) Fitting size is given in standard national pipe size (NPT).
4) Flow rate is given in gallons per minute. Use only as a guide.
5) Consult **Peerless** Engineering for the recommendations for your application.

SECTION 3-7
FLOUR GATE CONNECTION

WARNING A **flour gate** can cause personal injury or death. Shut off air supply and lock out before working on **flour gate** system.

Bulk flour loads into your **Peerless** mixer through the **flour gate** located on top of the canopy.

Using the **flour gate** prevents water and dough from entering the flexible connection or the flour hopper. The **flour gate** cannot be used to hold bulk flour in the flour hopper. A valve must be attached to the bottom of the flour hopper to hold the flour in the hopper. **Peerless** does not supply this valve.

Two types of **flour gates** are available for **Peerless** mixers: a standard **butterfly-style gate** and an optional **sliding gate**. See **Chapter 2, Options Checklist**, for your mixer's gate type.

NOTE: A **butterfly-style flour gate** requires a minimum of 10" between the top of the **flour gate** and the bottom of the flour hopper.

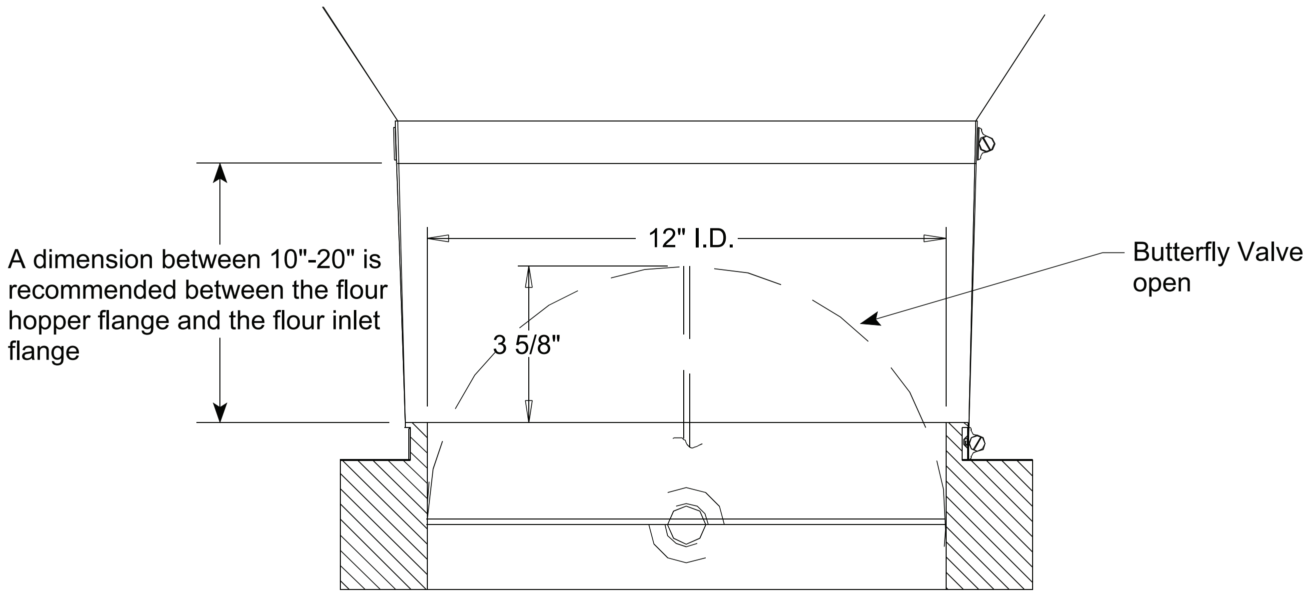
Connect the flour hopper to the mixer's **flour gate** with a flexible connection, or sleeve (not supplied). A flexible connection prevents any movement or vibration of the mixer from moving the flour hopper. If the flour hopper is mounted on load cells, the flexible connection is needed to get an accurate weight reading of the flour in the hopper.

Attach the flexible connection, or sleeve, to the mixer's **flour gate** by sliding it over the 13" (31 cm.) diameter shoulder of the flour flange plate. Use a hose clamp or band clamp (not supplied) to fasten the sleeve to the flour flange plate. See **Figure 3-7-1**.

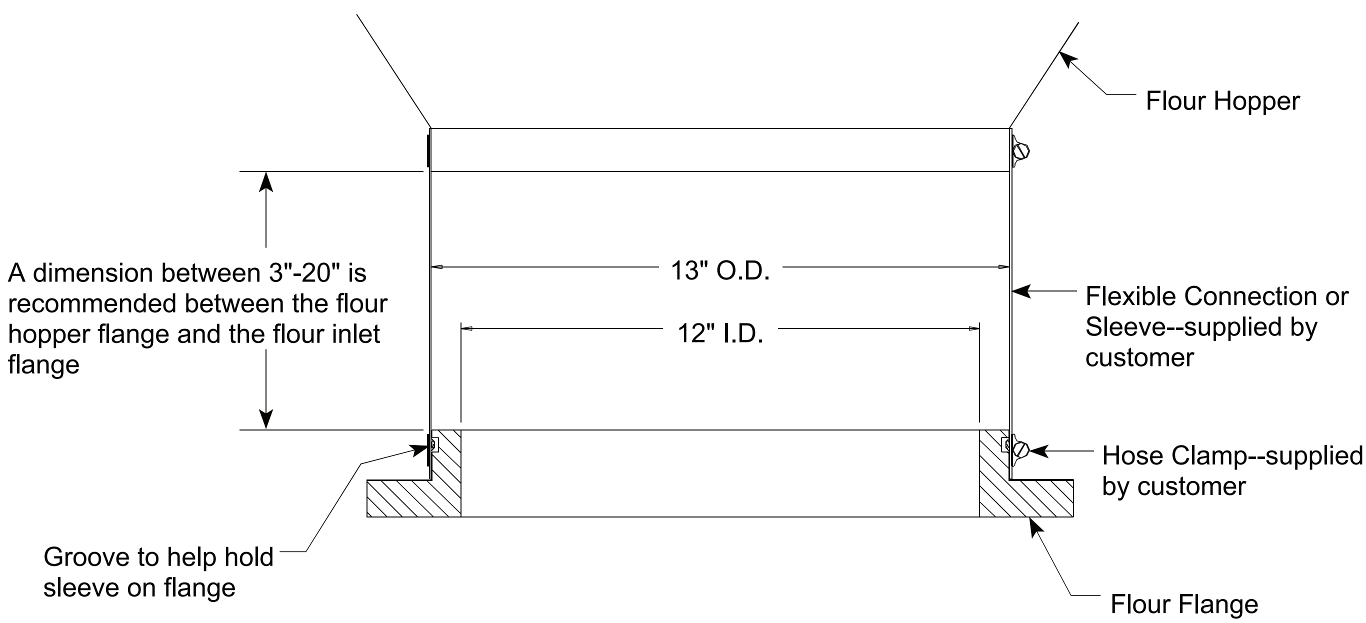
SECTION 3-7

FLOUR GATE CONNECTION FIGURE 3-7-1

STANDARD BUTTERFLY-STYLE GATE



OPTIONAL SLIDING GATE



SECTION 3-8
AIR SUPPLY CONNECTION

⚠	**CAUTION**	Pressurized air may cause personal injury. Air pressure lines must be shut off, locked out and remaining air removed from system before connecting to mixer.

Your **Peerless** mixer needs compressed air to operate. When connecting the compressed air supply to the mixer, use an air pressure of 80 PSI (5.5 bar) or more at 1.3 SCFM (.037m^3/min). Clean, dry air is best. Do not install a lubricator in the supply line to the mixer.

The **air supply connection** is located on the top of the mixer. The connection fitting is a 1/4" NPT pipe coupling. Remove the factory-installed pipe plug before connecting the compressed air supply line.

For air supply lines longer than 10 feet (3 m), we recommend a 3/8" pipe size. For less than 10 feet (3 m), a 1/4" pipe size can be used.

All **Peerless** standard mixers include a **filter-regulator** with shut-off valve. These parts are located behind the end doors on the left-hand end of the mixer.

SECTION 3-9
LIQUID INGREDIENT CONNECTIONS

Standard **Peerless** mixers are built with two fittings on the mixer canopy for loading liquid ingredients into the bowl. You decide the number of fittings needed and their locations when ordering your mixer. See the **Certified Data Sheet** for the quantity, type, size and location of **liquid ingredient connections** for your mixer.

Standard fittings are 2" bevel-seat sanitary fittings. Listed below are the vendor part numbers for fittings and mating parts.

LIQUID INGREDIENT FITTING VENDOR TABLE 3-9-1

FITTING	VENDOR		
	ROBERT JAMES	TRI-CLOVER	STA-RITE
Threaded Ferrule **(Mounted on Mixer)**	RJ15WL-2"	15WL-2"	BK29-93
Tygon hose Adapter **2" ID Hose**	RJ14AHT-2"	14AHT-2"	BK129-21
Hex Union Nut	RJ13H-2"	13H-2"	BK36-3
Rubber Hose Adapter **2" ID Hose**	RJ14AHR-2"	14AHR-2"	BK29-15
Ferrule (Short Welding)	RJ14PR-2"	14WL-2"	BK29-33
Cap	RJ16A-2"	16A-2"	BK3-3

SECTION 3-10
FLOUR DUST VENT

The **flour dust vent** allows displaced air to escape from the bowl during flour loading and mixing. The **flour dust vent** is located on the mixer canopy (bowl cover). The **flour dust vent** fitting is a 5 1/4" (13 cm.) diameter, 8" (20 cm.) high tube welded to the canopy. The cage, 24" (61 cm.) long, is covered by the dust vent sock and attached to the dust vent tube. The dust vent sock slips over the cage, and the excess sock material is tucked inside the bottom of the cage. A hose clamp secures the cage and sock to the dust vent tube.

Because of the height of the **flour dust vent**, it is shipped loose in the left-hand column and must be mounted on the mixer before operation.

Use the following steps and **Figure 3-10-1** to install the **flour dust vent**:

1. Remove the cage and sock from the mixer column.

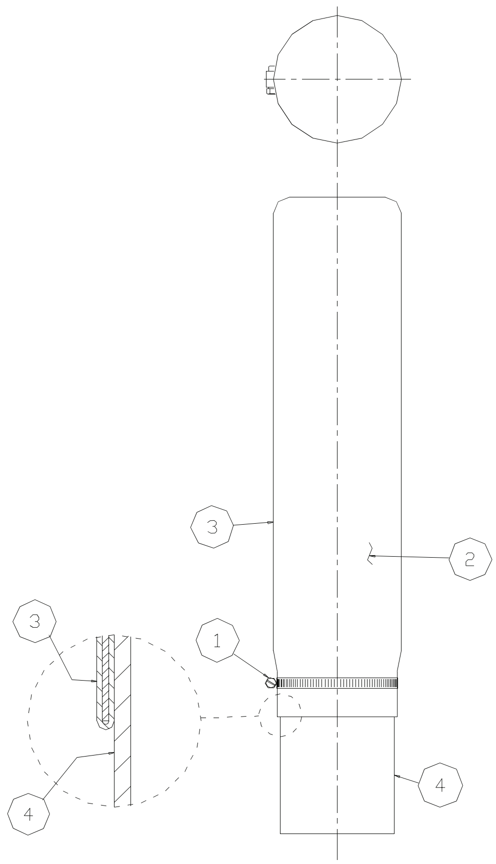
2. Place the cloth cover on the cage and tuck excess sock material inside the bottom of the cage.

3. Place the sock-covered cage on the dust vent tube.

4. Secure using hose clamps provided.

SECTION 3-10

FLOUR DUST VENT ASSEMBLY FIGURE 3-10-1



ITEM	QTY	DESCRIPTION
1	1	Hose Clamp
2	1	Cage Wire
3	1	Dust Vent Sock
4	1	Dust Vent Tube